Technical Report: Emotion Detection AI Training and Deployment Pipeline

1. Project Overview This section outlines the development and training pipeline for a YOLOv8s object detection model, designed for the Study Hive / ConnectEd platform. The model's primary objective is to perform real-time classroom monitoring by classifying student webcam feeds into three targeted states: Engaged, Confused, and Disengaged.

2. Data Acquisition and Preprocessing

- **Dataset Aggregation:** The foundational dataset, originally sourced from Roboflow, comprised 10 fine-grained emotion classes. This was augmented by merging four supplementary datasets, resulting in a finalized distribution of approximately 8,416 training images, 1,504 validation images, and 835 test images.
- **Label Consolidation:** To align with the platform's practical requirements, the original 10 classes were remapped into three semantically meaningful categories. The "Confused" class (derived from Worried and Fear labels) was designated as the key target metric due to its critical role in identifying struggling students.

3. Mitigation of Class Imbalance Following label consolidation, the dataset exhibited severe class imbalance, with the critical "Confused" class representing only 7.5% of the total data. This was addressed utilizing a three-tiered data augmentation strategy:

- **Targeted Augmentation:** Applied Albumentations using escalating intensity levels (Light, Medium, Heavy) to bring minority classes up to 80% of the majority class count.
- **Synthetic Data Generation:** Leveraged the Gemini 2.0 Flash API to generate synthetic, demographically diverse face portraits mapped to specific emotion prompts.
- **Advanced Style Transfer:** Utilized OpenCV-based transformations, including emotion-aware mood lighting and elastic distortion, to enhance the visual diversity of the training samples.

4. Training Methodology A two-phase training strategy was implemented to ensure stable convergence and prevent the degradation of pretrained weights:

- **Phase 1 (Warmup):** A 10-epoch initial phase utilizing a frozen CSPDarknet backbone, allowing the detection head to adapt to the new emotion classes.
- **Phase 2 (Full Fine-Tuning):** A 150-epoch phase with all layers unfrozen, employing an AdamW optimizer, cosine annealing, and comprehensive online augmentations (e.g., Mosaic, MixUp, Random Erasing). Regularization techniques, including label smoothing and dropout, were applied to prevent overfitting.

5. Evaluation and Deployment Architecture

- **Model Evaluation:** Performance was validated using standard object detection metrics (mAP50, mAP50-95, precision, and recall), supplemented by a specialized analysis block dedicated to optimizing confidence thresholds for the "Confused" class.
- **Real-Time Inference:** The finalized model was deployed via a Flask server utilizing WebSockets to process base64 webcam feeds from the browser.

- **Pipeline Features:** The deployment pipeline incorporates temporal smoothing to mitigate classification flickering, position-based tracking, and an aggregate "Engagement Score" that triggers automated visual alerts for educators when disengagement thresholds are met.

Summary: Best Results Across All Runs

Run	Epochs Trained	Best mAP50	Best mAP50-95	Best Precision	Best Recall
v1 Phase 1	10	0.5994	0.4992	0.4901	0.8040
v1 Phase 2	9 (early stop)	0.6516	0.5494	0.5996	0.8744
v2 Phase 1	10	0.5444	0.4246	0.4502	0.7876
v2 Phase 2	45 (early stop)	**0.6477**	**0.5491**	0.5279	0.7988
v3 Phase A	35	**0.7299**	**0.6301**	**0.5900**	**0.8131**

> IMPORTANT
> **v3 is your best model** with **mAP50 = 0.730** and **mAP50-95 = 0.630** — a significant improvement over v1 and v2.

Run-by-Run Breakdown

◆ v1 — Phase 1 (Frozen Backbone, 10 epochs)

The warmup phase adapted the detection head to the 3 emotion classes while keeping the CSPDarknet backbone frozen.

Epoch	Box Loss	Cls Loss	Precision	Recall	mAP50	mAP50-95
1	0.8752	1.2631	0.3895	0.8040	0.4607	0.3568
5	0.6143	0.8790	0.4514	0.7299	0.5390	0.4566
10	**0.5157**	**0.7372**	**0.4857**	**0.7241**	**0.5994**	**0.4992**

◆ v1 — Phase 2 (Full Fine-Tuning, 9 epochs then stopped)

Unfroze all layers. Only ran 9 epochs (likely stopped manually or hit an issue).

Epoch	Box Loss	Cls Loss	Precision	Recall	mAP50	mAP50-95
1	1.3158	1.5677	0.3684	**0.8744**	0.4768	0.3658
5	0.9012	1.1560	0.4824	0.7963	0.5976	0.4623
9	**0.8111**	**1.0675**	**0.5996**	**0.6777**	**0.6463**	**0.5494**

Key observation: Precision improved significantly (0.37 → 0.60), but recall dropped (0.87 → 0.68) — the model became more selective. mAP50-95 peaked at 0.549.

◆ v2 — Phase 1 (Frozen Backbone, 10 epochs)

Re-trained after additional data augmentation. Slightly more data.

Epoch	Box Loss	Cls Loss	Precision	Recall	mAP50	mAP50-95
1	0.8254	1.2691	0.3612	0.7839	0.4196	0.3126
5	0.5901	0.9551	0.4086	0.7663	0.5040	0.3824
10	**0.5036**	**0.8458**	**0.4502**	**0.7876**	**0.5444**	**0.4246**

Key observation: Lower val cls_loss (0.846 vs 0.737 in v1) — more data but harder classification due to augmented images.

◆ v2 — Phase 2 (Full Fine-Tuning, 45 epochs then early stopped)

Longest fine-tuning run. Early stopping triggered at epoch 45 (patience = 30).

Epoch	Box Loss	Cls Loss	Precision	Recall	mAP50	mAP50-95
1	1.1885	1.5629	0.3354	0.8076	0.3916	0.2470
10	0.7519	1.1418	0.4559	0.7677	0.5628	0.4525
20	0.6717	1.0494	0.5047	0.7701	0.5959	0.4925
30	0.6342	0.9940	0.5025	0.7831	0.6175	0.5172
36	0.6154	0.9724	0.5108	**0.8173**	0.6390	0.5410
40	0.6096	0.9600	0.5116	0.8151	0.6420	0.5400
44	**0.5974**	**0.9432**	**0.5279**	**0.7988**	**0.6477**	**0.5491**

Key observation: Steady improvement over 45 epochs. Best mAP50 = 0.648, close to v1 Phase 2 but with better recall (0.80 vs 0.68). More balanced model.

◆ v3 — Phase A (Best Run, 35 epochs)

Your **best-performing run**. Started from a stronger checkpoint. Trains showed significantly higher metrics from the start.

Epoch	Box Loss	Cls Loss	Precision	Recall	mAP50	mAP50-95
1	0.6579	0.9784	0.4664	0.7182	0.5861	0.4646
5	0.5647	0.8678	0.5689	0.7906	0.7124	0.6100
10	0.5434	0.8403	0.5701	0.7625	0.7047	0.6079
13	0.5348	0.8267	**0.5876**	**0.7895**	**0.7299**	**0.6199**
20	0.5160	0.7970	0.5800	0.7953	0.7210	0.6165
25	0.5019	0.7725	0.5851	0.7975	0.7293	0.6287
30	0.4923	0.7654	0.5753	0.8104	0.7277	**0.6301**
33	**0.3592**	**0.5258**	**0.5900**	**0.7904**	**0.7278**	**0.6306**
35	0.3539	0.5185	0.5874	0.7904	0.7267	0.6292

> **NOTE**
> At epoch 31, there's a dramatic drop in train losses (box: 0.49 → 0.37, cls: 0.77 → 0.55). This coincides with **Close Mosaic** kicking in at epoch 31 (last 15 of a likely 45-epoch config), which disables mosaic augmentation and lets the model focus on clean single-image learning.

```
mAP50-95 Progression:

v1 Phase 1   0.499
v1 Phase 2   0.549
v2 Phase 1   0.425
v2 Phase 2   0.549
v3 Phase A   0.630  ← BEST
```

```
mAP50 Progression:

v1 Phase 1   0.599
v1 Phase 2   0.652
v2 Phase 1   0.544
v2 Phase 2   0.648
v3 Phase A   0.730  ← BEST
```

Pipeline Order

```
1. merge_datasets.py      → Combine all raw datasets
2. remap_labels.py        → 10 → 3 class remapping
3. count_labels.py        → Check distribution
4. augment_minority_classes.py  → Fix imbalance with Albumentations
5. generate_synthetic_faces.py  → Generate new faces via Gemini API
6. style_transfer_augment.py    → Advanced face-specific augmentation
7. train_model.py         → Two-phase YOLOv8 training
8. evaluate_model.py      → Test set evaluation
9. export_model.py        → Export for deployment
10. live_inference.py      → Real-time inference server
```